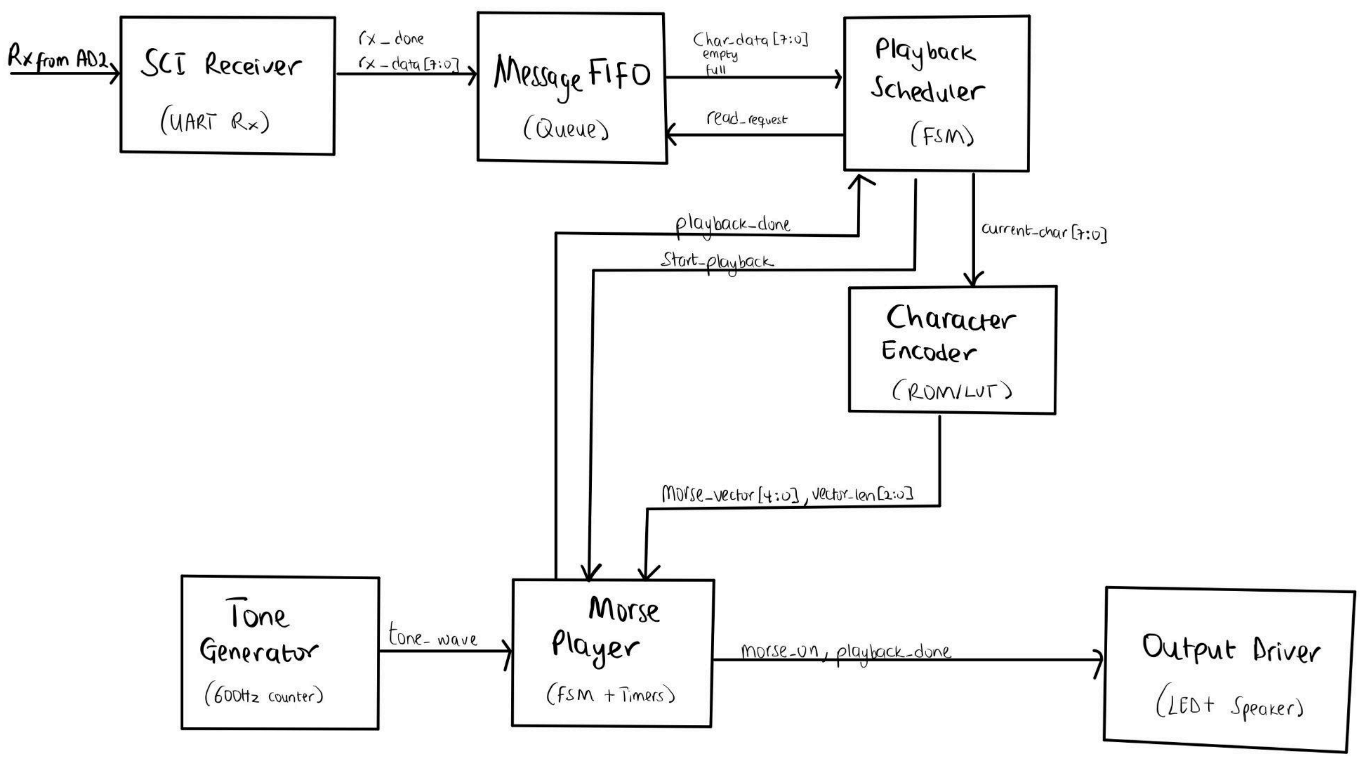
ENGS 31 FINAL PROJECT

MORSE CODE CONVERTER

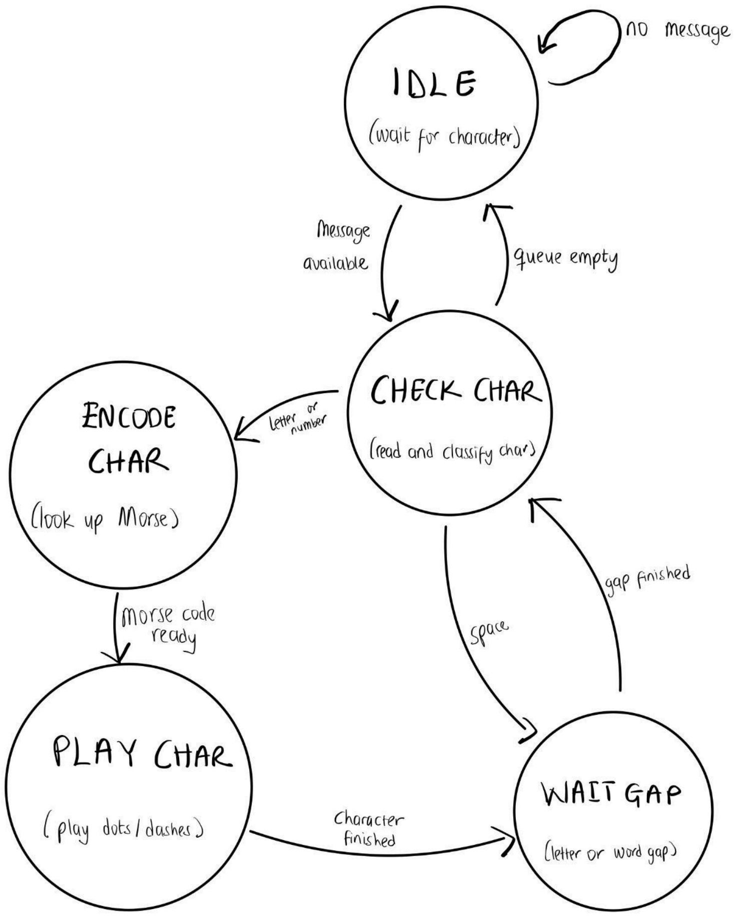
PAPA YAW OWUSU NTI, GENT MAKSUTAJ

Building intuition:

BLOCK DIAGRAM



HIGH LEVEL STATE MACHINE



DATAPATH

FINITE STATE MACHINE

1. SCI Receiver FSM(watch the rx line for a start bit, then sample the 8 data bits at the right times, then check the stop bit, then pulse rx_done)

    IDLE waits for the serial line to go low. UART rests at Rx = 1, so a low value means a possible start bit.

    START_WAIT waits half a bit time. This lets us check the middle of the start bit. If Rx is still 0, it is a real start bit. If it went back to 1, it was probably noise.

    DATA_WAIT waits one full baud period between samples.

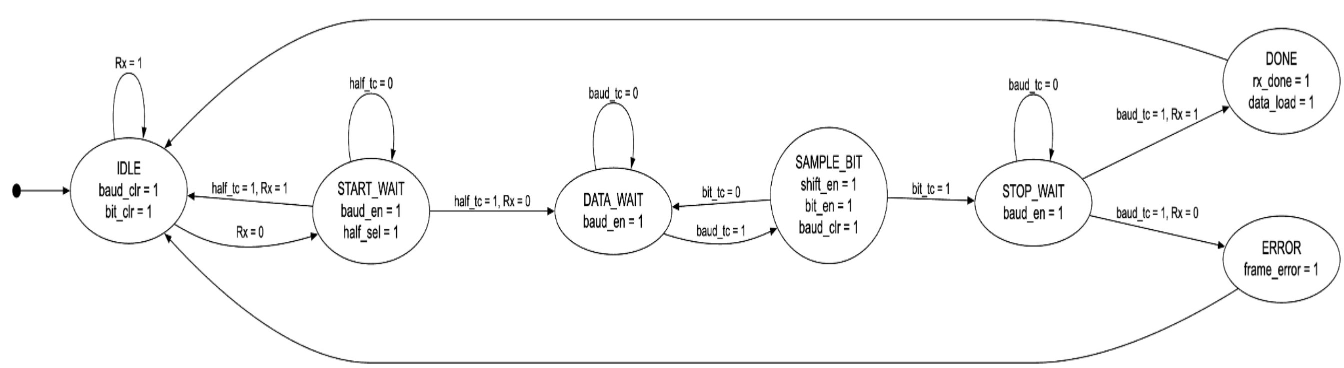
    SAMPLE_BIT is the Moore state where the actual data bit is shifted into the receive shift register. It also increments the bit counter.

    STOP_WAIT waits one full baud period, then checks the stop bit. Stop bit should be 1.

DONE gives a one-cycle rx_done pulse and loads the completed byte into rx_data.
ERROR gives a one-cycle frame_error pulse if the stop bit was bad.

Inputs:
Rx = serial input line from the AD2
half_tc = half-bit timer terminal count
baud_tc = full-bit timer terminal count
bit_tc = bit counter terminal count after 8 data bits

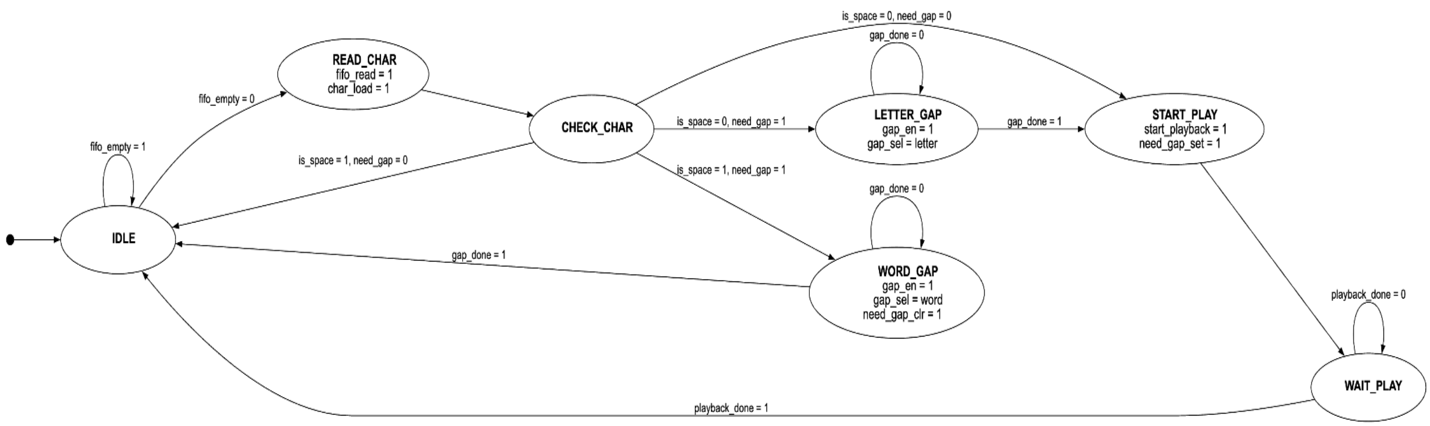
Outputs:
baud_clr = clears the baud timer
baud_en = enables the baud timer
half_sel = makes the baud timer count half a bit period
bit_clr = clears the bit counter
bit_en = increments the bit counter
shift_en = shifts the sampled Rx bit into the receive shift register
data_load = loads the completed byte into rx_data
rx_done = pulses high when a valid byte is ready
frame_error = pulses high when the stop bit is invalid



2. Playback Scheduler FSM(reads characters from the FIFO, decides whether each character is a space or a playable symbol, inserts the correct letter/word gap, and tells the Morse Player when to start.)
   Inputs:
   fifo_empty = no character is waiting in the FIFO
   is_space = current character is a space

need_gap   = a previous character was played, so a gap may be needed
gap_done   = letter/word gap timer is finished
playback_done = Morse Player finished the current character

Outputs:
fifo_read = read one character from the FIFO
char_load = store FIFO output into current_char register
gap_en = enable the gap timer
gap_sel = choose letter gap or word gap
need_gap_clr = clear the need_gap flag after a word gap
start_playback = tell Morse Player to begin playing current character
need_gap_set = remember that a character has been started/played



3. Morse Player FSM(plays one Morse-encoded character. It loads the Morse pattern, checks each symbol, turns morse_on high for a dot or dash duration, inserts 1T gaps between symbols, and pulses playback_done when the character is finished.)
   Inputs:
   start_playback = Scheduler tells the Morse Player to begin
   unit_done   = one T-unit has finished
   bit_done   = all valid waveform bits have been played
   length_zero   = loaded character has no playable Morse waveform
   code_bit   = current MSB of the 19-bit Morse shift register

   Outputs:

code_load = load morse_data into the 19-bit code register

length_load = load length_data into the 5-bit length register

code_shift = shift the Morse code register left by one bit

bit_clr = clear the bit counter

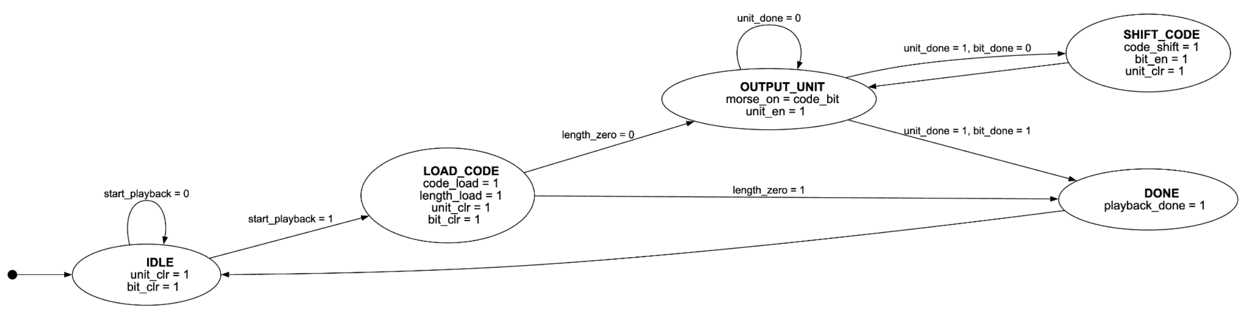
bit_en = increment the bit counter

unit_clr = clear the T-unit timer

unit_en = enable the T-unit timer

morse_on = current Morse output bit; drives LED/speaker gate

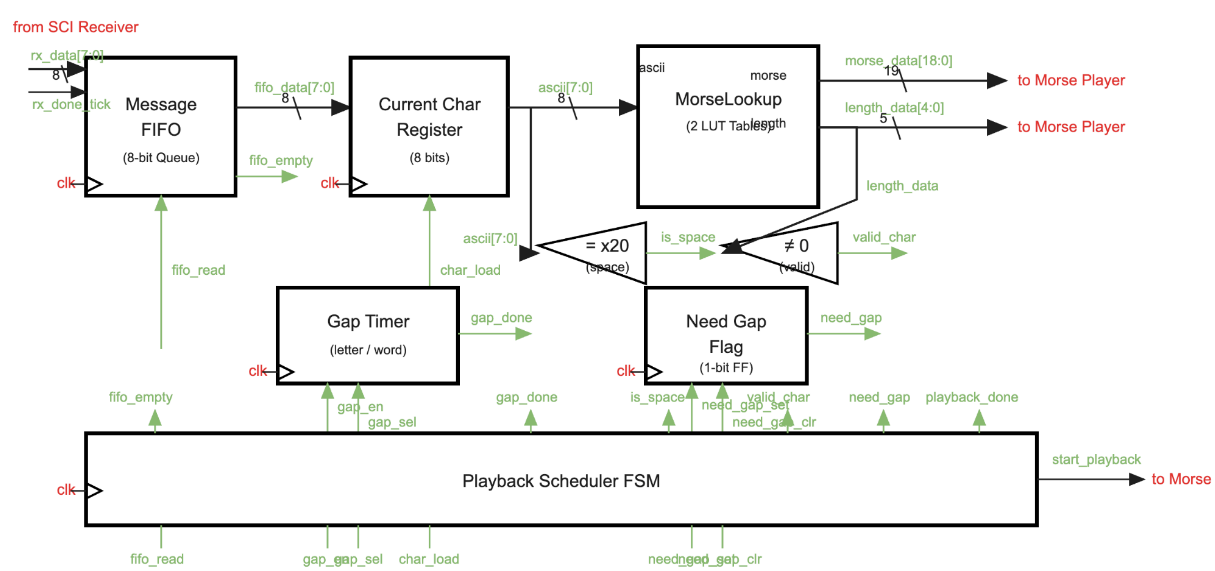
playback_done = current character is finished



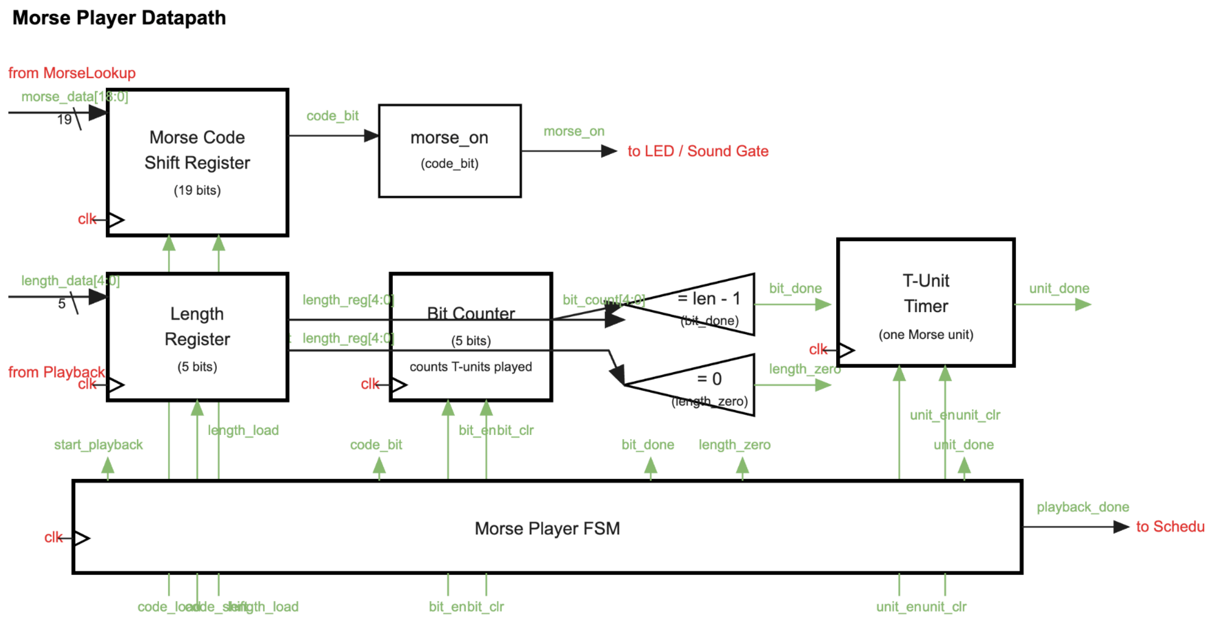
RTL DATAPATH.

1. Playback Scheduler Datapath

Playback Scheduler Datapath



2. Morse Player Datapath



Actual Toplevel datapath